

SQL Standards & Core Concepts

Q. What is SQL Discuss its features, the importance of SQL Standards, and categorize the different types of SQL commands with examples.

[DEF] Definition to Remember

SQL (Structured Query Language) is the standard declarative language for communicating with Relational Database Management Systems (RDBMS). It allows users to create, manipulate, manage, and query databases. SQL is governed by ANSI and ISO standards to ensure cross-platform compatibility.

1. SQL Standards

Standardizing bodies ANSI and ISO define the SQL standard to ensure compatibility across different database vendors (Oracle, MySQL, SQL Server).

Standard	Key Feature
SQL-86	First official standard
SQL-92	Introduced Joins and expanded query ca
SQL:1999 (SQL3)	Introduced object-relational features,

Importance: Ensures core SQL commands work consistently across databases, preventing vendor lock-in.

2. Features of SQL

- * Declarative Language: Users specify *what* data they want, not *how* to retrieve it.

* Ea

3. Types of SQL Commands

Category	Full Form	Purpose	Key Commands
DDL	Data Definition Language	Define/modify database structure	`CREATE`, `ALTER`, `DROP`, `TRUNCATE`,
DML	Data Manipulation Language	Insert, update, delete, retrieve data	`SELECT`, `INSERT`, `UPDATE`, `DELETE`
DCL	Data Control Language	Manage user access & permissions	`GRANT`, `REVOKE`
TCL	Transaction Control Language	Manage transactions & data integrity	`COMMIT`, `ROLLBACK`, `SAVEPOINT`

Examples:

```
-- DDL
CREATE
-- DML
INSERT
-- DCL
GRANT
-- TCL
COMMIT
```

[MUST-WRITE] Must-Write Points (for 10 marks)

1. SQL = Structured Query Language; standard language for RDBMS communication.
2. Governed by ANSI and ISO standards (SQL-86, SQL-92, SQL:1999).
3. SQL is declarative -- specifies *what* to retrieve, not *how*.
4. DDL: structure operations (CREATE, ALTER, DROP, TRUNCATE, RENAME).
5. DML: data operations (SELECT, INSERT, UPDATE, DELETE).

6. DCL: access control (GRANT, REVOKE).

7. TCL: transaction management (COMMIT, ROLLBACK, SAVEPOINT).

[RECALL] Quick Recall

SQL -> ANSI/ISO Standards (SQL-86, SQL-92, SQL:1999) -> Declarative -> DDL (structure) -> DML (data) -> DCL (access) -> TCL (transactions)